8th Grade Digital Discoveries

Family Overview · Optima Academy Online

~4¼ hrs per week · ~50 min/day, M–F

About this overview. This document is an example of how the course might be taught. More topics, tools, and projects are approved for this course than appear here, so scholars may see some variation in topics, tools, and projects from course to course, and between the live school and On-Demand. Each teacher also leans into their own strengths and passions; the structure, expectations, and time commitment below are representative of the course as designed.

ABOUT THIS COURSE

A capstone year of middle-school computing. Scholars take on digital identity, networks and privacy, emerging technologies, programming, data and simulation, and end the year by designing software using a real design-thinking workflow.

THE YEAR AT A GLANCE

Unit 1 — Digital Identity & Communication ~6 weeks
Online presence, audience, and responsible communication.

Unit 2 — Networks, Privacy & Digital Wellness ~6 weeks
How networks work, privacy trade-offs, and healthy screen habits.

Unit 3 — Emerging Technologies & Society ~6 weeks
AI, automation, and how new tech reshapes work and culture.

Unit 4 — Programming & Computational Thinking ~6 weeks
Algorithms, logic, and coding in Python.

Unit 5 — Data, Science & Simulation ~6 weeks
Data analysis and using simulations to model the world.

Unit 6 — Software Development & Design Thinking ~6 weeks
Design-thinking workflow culminating in a build-and-pitch project.

Capstone & projects. The Unit 6 software-design project is the course capstone.

A TYPICAL WEEK

- One lesson page anchoring the week's topic.
- One activity or assignment that applies it (often hands-on).
- One quiz or knowledge check.
- Periodic VR experiences inside Optima's VR campus.

TIME COMMITMENT

250 minutes per week (master schedule)

- Lesson reading & video: ~15 min/day
- Assignment / build work: ~25 min/day
- Quiz, reflection, or VR session: ~10 min/day

Project and capstone weeks may run longer.

WHAT SCHOLARS NEED

- Optima's VR headset (provided) for VR experiences
- Microsoft 365 / OneDrive account
- Microphone for voice-based work
- Reliable internet

AI in this course. The course treats AI as a tool to understand and critique — scholars learn what AI does well and where human judgment matters.

On-Demand vs. Live. On-Demand (asynchronous): self-paced with due dates for pacing; a teacher is available for support, feedback, and grading. **Live school:** live-school scholars take the same course in Optima's VR campus with a teacher and classmates (daily for core courses).